

Mini FRD - [Timesheet] Timer – Update Calculation Logic to Daily Total Cap

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1.0	@17 avril 2026	Thảo Thanh	

1. Overview

Context: The current Timer limits 8 hours independently per task. New requirement: The Timer must check the **total time already logged today** before starting a new session.

Formula:

```
remaining_capacity = 480 minutes - total_logged_today
```

In Scope

- Update limit-check logic: from per-task to daily total
- Display soft-limit warning (⚠️) when exceeding the [daily_cap]-hour threshold — **does not block**
- Smart Auto-fill in the Review Modal when stopping the Timer

- Warning when a Manual Log entry exceeds the threshold
- Handling sessions that cross midnight (cross-midnight)

Out of Scope

- Changes to the manual worklog flow (non-timer) — except the over-cap warning
- Editing historical worklog data prior to the release date

2. Functional Requirements

FR-01 – Core Rule

The system must aggregate the total time logged today (including all completed and active worklogs), compare it against the standard threshold, and display this information when the user opens the Log Work form.

- The formula is calculated at: (1) when the Log Work form is opened (Manual or End Timer) and (2) in the Review Modal when stopping the Timer
- **Not used to block** — for informational and suggestive purposes only

FR-02 – Display Remaining Capacity When Opening the Log Work Form

Trigger: User opens the Log Work form — including (a) Manual Log or (b) clicking End Timer.

>  **Do not display while the Timer is running.**

Behavior:

Condition	System Behavior
<code>remaining_capacity > 0</code>	Display below Duration: <i>"You have [Xh Ym] of standard time left to log today."</i> Ceiling = remaining_capacity
<code>remaining_capacity ≤ 0</code>	Timer still allows start. Display below Duration: <i>"You have reached today's [daily_cap]-hour standard threshold. You are Extra/OT by [Xh Ym]."</i> Ceiling = 0

FR-03 – Over-Limit Warning and Icon

Trigger: User opens the Log Work form (any flow).

Behavior:

- Notice is placed **below the Duration input field**, not inline
- When threshold has not been exceeded: normal color — *"You have [Xh Ym] of standard time left to log today."*
- When 8 hours have been reached or exceeded:
 - Color changes to **orange/yellow**
 - Display  icon + *"You have reached today's [daily_cap]-hour standard threshold. You are Extra/OT by [Xh Ym]."*

FR-04 – Review Modal with Smart Auto-Fill (Stop Timer Flow)

Trigger: User clicks Stop → system opens the Log Work form (Review Modal).

> The system **does not auto-save** — the user must review and submit manually.

Auto-Fill Logic:

- Duration is automatically filled with the **actual elapsed time of the session** (only if the user has not edited the field)
- Below Duration, display: *"Actual logged time: Xh Ym. You are Extra/OT by Zh Wm."*
- If the user manually enters a value exceeding the threshold → do not re-auto-fill, only display: *"You are Extra/OT by [Xh Ym] compared to today's standard hours."*
- Submit is always allowed — does not block

Concrete example:



User has logged 7h. Timer currently running for 10h.

→ Auto-fill: 10h (actual elapsed time)

→ Below Duration: "Actual logged time: 10h. You are Extra/OT by 9h."

→ User does not edit → submits 10h normally

→ User manually enters "12h" → displays: "You are Extra/OT by 11h." → Submit normally

FR-05 – Manual Log: Warning When Entry Exceeds the Limit

Trigger: User opens the Log Work form manually (not via Timer).

Behavior:

- When the form opens and the user has not yet interacted: auto-fill with `remaining_capacity` **at the time the form is opened** (current snapshot)
- If the user enters a value that would make the daily total exceed 480 minutes → display below Duration: *"If saved, today's total will be [X]h — you are Extra/OT by [Yh Ym] compared to standard hours."*
- Submit is always allowed — does not block

💡 **Key difference between FR-04 and FR-05:**

- **FR-04 (Stop Timer):** auto-fill = **Actual elapsed time** of the session
- **FR-05 (Manual Log):** auto-fill = `remaining_capacity` at the *time the form is opened* (current snapshot)

FR-06 – Server-Side Daily Capacity Query

Trigger: Any request that creates or updates a worklog record.

- The system aggregates the total time logged today (including the entry being saved) to update the remaining capacity returned to the UI
- Response includes: `total_logged_today`, `daily_cap` (480 minutes), `remaining_capacity`
- **Not used to block or reject submissions**
- Frontend must include `timezone_offset` (e.g., `+07:00`) or `local_timestamp` in every API request, so the Backend can correctly map the user's local day boundary when aggregating `total_logged_today`.

FR-07 – Remaining Daily Capacity Data

Trigger: Any time the user interacts with the Timer.

Data returned:

Field	Value
<code>total_logged_today</code>	Total time logged today
<code>daily_cap</code>	480 minutes
<code>remaining</code>	<code>daily_cap - total_logged_today</code>

FR-08 – Cross-Midnight Session Handling

Scenario: Timer starts before midnight (e.g., 23:45) and stops after midnight (e.g., 00:20).

The session must be split at the midnight boundary (based on the user's timezone):

- **Record 1:** Assigned to the start date. Duration calculated from session start → 23:59:59
- **Record 2:** Assigned to the following day. Duration calculated from 00:00:00 → the stop time
- Each record is counted independently toward the daily total of its respective day

3. Business Logic Flow

General note: All notices and business logic **are only triggered when the user opens the Log Work form** — including (a) ending the Timer or (b) opening a Manual Log. Not displayed while the Timer is running. The system **does not block** any action.

3.1 End Timer Flow

Trigger: User clicks Stop → system opens the Log Work form (Review Modal). No auto-save.

When the form opens, the system calculates:

```
remaining_capacity = 480 - total_logged_today
ceiling = remaining_capacity (recorded at the time the session started)
```

The Duration field is automatically filled with the ceiling (only if the user has not edited the field).

Case 1 — remaining > 0

Element	Behavior
Auto-fill	Filled with actual elapsed time of the session
Notice below Duration	"Actual logged time: Xh Ym. Remaining today: Zh Wm." — normal color ($khi\ actual \leq daily_cap$) "Actual logged time: Xh Ym. You are Extra/OT by Zh Wm." — orange/yellow color ($khi\ actual > daily_cap$)
Extra/OT button	Not highlighted
Submit	Allowed normally

Case 1 — actual > daily_cap (Stop Timer flow, updated)

Log Work
XCOR-2281

☐ Extra Time
 ☐ Over Time
 ☒ Time Info
 ☐ Sync to Jira

Start Date: Apr 22, 2026
 End Date: Apr 22, 2026

Start Time: 22:00
 End Time: 08:00

Type: Type
 Duration: 10h AUTO-FILL

⚠ Actual logged time: **10h**. You are Extra/OT by **9h** compared to today's standard hours.
Please activate the Extra/OT button to record your overtime.

Description:

● 7h logged today
Discard
Submit

Scenario: 7h already logged. Timer ran 10h. Auto-fill = 10h (actual). Warning shown, submit not blocked.

Case 2 — remaining = 0

Element	Behavior
Auto-fill	Filled with 0 (ceiling = 0)

Element	Behavior
Notice below Duration	⚠️ + "You have reached today's [daily_cap]-hour standard threshold. You are Extra/OT by 0h 0m." — orange/yellow color
Extra/OT button	Highlighted
Submit	Allowed normally

Log Work

XCOR-2281

☐ Extra Time
☐ Over Time
☒ Time Info
☐ Sync to Jira

Start Date

Apr 22, 2026

End Date

Apr 22, 2026

Start Time

05:00 CH

End Time

06:00 CH

Type

Type

Duration

0h

CEILING = 0

⚠️ You have reached today's 8-hour standard threshold. You are Extra/OT by 0h 0m.
Hover ⬆️ — "You are in the Extra/OT zone. Please activate the Extra/OT button."

Description

• 8h logged today

Discard

Submit

Case 3 — remaining < 0

Element	Behavior
Auto-fill	Filled with 0 (ceiling = 0)
Notice below Duration	⚠️ + "You have reached today's [daily_cap]-hour standard threshold. You are Extra/OT by [Xh Ym]." — orange/yellow color

Element	Behavior
Hover 	<i>"You are in the Extra/OT zone. Please activate the Extra/OT button to record your overtime."</i>
Extra/OT button	Highlighted
Submit	Allowed normally — not rejected

Log Work

XCOR-2281

Extra Time

Over Time

Time Info

Sync to Jira

Start Date

Apr 22, 2026

End Date

Apr 22, 2026

Start Time

10:00 SA

End Time

08:00 CH

Type

Type

Duration

0h

CEILING = 0

You have reached today's 8-hour standard threshold. You are Extra/OT by 1h.

Hover

"Please activate the Extra/OT button to record your overtime."

Description

9h logged today

Discard

Submit

Note: If the user manually edits Duration to exceed the threshold → do not re-auto-fill, only display: *"You are Extra/OT by [Xh Ym] compared to today's standard hours."*

3.2 Manual Log Work Flow

Trigger: User opens the Log Work form manually (not via Timer).

When the form opens, the system calculates `remaining_capacity` **at the time the form is opened** (current snapshot — no session start to reference).

The Duration field is automatically filled with the current `remaining_capacity` (only if the user has not edited the field).

Case 1 — `remaining > 0`

Element	Behavior
Auto-fill	Filled with <code>remaining_capacity</code> at the time the form is opened
Notice below Duration	<i>"You have [Xh Ym] of standard time left to log today."</i> — normal color
Extra/OT button	Not highlighted
Submit	Allowed normally

Log Work

XCOR-2281

☐ Extra Time

☐ Over Time

☒ Time Info

☐ Sync to Jira

Start Date

Apr 22, 2026

End Date

Apr 22, 2026

Start Time

05:34 CH

End Time

05:34 CH

Type

Type

Duration

1h

AUTO-FILL

You have 1h 0m of standard time left to log today.

Description

7h logged today

Discard

Submit

Case 2 — remaining = 0

Element	Behavior
Auto-fill	Filled with 0
Notice below Duration	 + "You have reached today's [daily_cap]-hour standard threshold. You are Extra/OT by 0h 0m." — orange/yellow color
Extra/OT button	Highlighted
Submit	Allowed normally

Log Work

XCOR-2281

☐ Extra Time
 ☐ Over Time
 ☒ Time Info
 ☐ Sync to Jira

Start Date

Apr 22, 2026

End Date

Apr 22, 2026

Start Time

05:34 CH

End Time

05:34 CH

Type

Type

Duration

0h CEILING = 0

 You have reached today's 8-hour standard threshold. You are Extra/OT by 0h 0m.
 Hover  → "Please activate the Extra/OT button to record your overtime."

Description

8h logged today

Discard

Submit

Case 3 — remaining < 0

Element	Behavior
Auto-fill	Filled with 0

Element	Behavior
Notice below Duration	⚠️ + "You have reached today's [daily_cap]-hour standard threshold. You are Extra/OT by [Xh Ym]." — orange/yellow color
Hover ⚠️ tooltip	"You are in the Extra/OT zone. Please activate the Extra/OT button to record your overtime."
Extra/OT button	Highlighted
Submit	Allowed normally — not rejected

Log Work

XCOR-2281

☐ Extra Time
 ☒ Over Time
 ☒ Time Info
 ☐ Sync to Jira

Start Date

Apr 22, 2026

End Date

Apr 22, 2026

Start Time

10:00 SA

End Time

08:00 CH

Type

Type

Duration

3h

USER EDITED

⚠️ You are Extra/OT by 2h compared to today's standard hours.

No re-auto-fill. Submit still proceeds normally.

Description

7h logged today

Discard

Submit

Note: If the user manually enters a Duration that exceeds the threshold → do not re-auto-fill, only display: "If saved, today's total will be [X]h — you are Extra/OT by [Yh Ym] compared to standard hours."

3.3 Key Differences Between End Timer and Manual Log

Difference	End Timer (FR-04)	Manual Log (FR-05)
Auto-fill source	Ceiling recorded at session start (fixed for the duration of the session)	<code>remaining_capacity</code> at the time the form is opened (current snapshot)
Additional display line	<i>"Actual logged time: Xh Ym. Remaining today: Zh Wm."</i> — hoặc <i>"You are Extra/OT by Zh Wm."</i> tùy trạng thái	This line is not displayed
Reason	Timer has a session start to reference	Manual Log has no session start

3.4 Post-Submit Flow (After submitting from any flow)

Step	Description
1	Record is committed to the DB
2	System re-aggregates <code>total_logged_today</code> and recalculates the new <code>remaining_capacity</code>
3	UI updates to display the latest remaining capacity
4 (if threshold exceeded)	Notice below Duration: <i>"You have reached today's [daily_cap]-hour standard threshold. You are Extra/OT by [Xh Ym]."</i> — not rejected, not blocked

4. Key Test Cases

TC ID	Scenario	Pre-condition	Steps	Expected Result
TC-01	End Timer: new day, nothing logged yet	0 minutes logged	Click End Timer → form opens	Ceiling = 8h. Below: <i>"8h remaining."</i> No warning.
TC-02	End Timer: partially logged	5h logged	Click End Timer → form opens	Ceiling = 3h. Below: <i>"3h remaining."</i>
TC-03	Log Work: daily cap already reached	Exactly 8h logged	Open Log Work form	⚠ + <i>"Extra/OT by 0h 0m."</i> Auto-fill = 0. Extra/OT highlighted.
TC-04	Log Work: total log exceeds ceiling when form opens	7h logged. Timer running for 2h then stopped.	Timer → Stop → form opens	⚠ + <i>"Extra/OT by 1h."</i> Hover ⚠: tooltip displayed.

TC ID	Scenario	Pre-condition	Steps	Expected Result
TC-05a	Stop Timer: user does not edit auto-fill	7h logged. Timer ran for 3h.	Stop → form opens → Submit default value	Auto-fill 1h. <i>"Actual: 3h. Remaining: 1h."</i> Submit → saves 1h.
TC-05b	Stop Timer: user manually enters over-threshold value	7h logged. Timer ran for 3h.	Stop → clear auto-fill → enter "3h" → Submit	No re-auto-fill. <i>"Extra/OT by 2h."</i> Extra/OT highlighted. Submit 3h normally.
TC-06	Manual Log: form opens for the first time, user has not entered anything	7h logged	Open Log Work form, make no changes	Auto-fill <i>remaining_capacity</i> = 1h.
TC-06b	Manual Log: user enters over-limit value	7h logged	Open form → clear auto-fill → enter "3h"	No re-auto-fill. <i>"Today's total = 10h — Extra/OT by 2h."</i> Submit normally.
TC-07	Concurrent sessions – 2 tabs	User opens 2 tabs, each logs 5h	Both submit	Both submissions succeed (serialized via DB lock). Daily total = 10h. "Extra/OT by 2h." (<i>Backend must implement Optimistic Locking or equivalent to prevent race conditions on <i>total_logged_today</i></i>)
TC-08	Cross-midnight session	Timer starts at 23:50	Session continues until 00:10 the next day	2 records: 10 minutes assigned to Day 1, 10 minutes assigned to Day 2.
TC-09	Timezone determines day boundary	User timezone = UTC+7, server = UTC	User starts/stops Timer near server midnight	Worklog date assigned per UTC+7, not UTC.

5. Assumptions

- Default daily cap = **480 minutes (8 hours)** — this is used as an **illustrative example** throughout this document only. In practice, the standard daily time threshold is a **configurable value** per user or per organization. All references to "8 hours" or "480 minutes" in this document reflect logic relative to the configured value, not a hardcoded constant.
- "Today" is determined by the **user's configured timezone**
- Both completed and active timer sessions are counted toward the daily total
- Manually entered worklogs (not via Timer) **are also counted** toward the daily total
- No historical worklogs (prior to the release date) will be changed or affected by the new logic
- Remaining capacity is calculated **once at session start** and used as the fixed ceiling for that session
- Sessions with an elapsed duration of 0 **are not saved** as worklog entries